

Missionaries and Cannibals Problem

Aim

To solve the **Missionaries and Cannibals Problem** using Python by safely transporting all missionaries and cannibals across the river without leaving missionaries outnumbered.

Algorithm

1. Represent the state as (**Missionaries Left, Cannibals Left, Boat Position**).
2. Start with all missionaries and cannibals on the left bank.
3. Generate all possible valid boat moves.
4. Check each state to ensure missionaries are never outnumbered by cannibals on either bank.
5. Use **Breadth-First Search (BFS)** to explore valid states.
6. Stop when all missionaries and cannibals reach the right bank.

Python Code

```
from collections import deque
```

```
def is_valid(m_left, c_left):
```

```
    m_right = 3 - m_left
```

```
    c_right = 3 - c_left
```

```
    if m_left < 0 or c_left < 0 or m_left > 3 or c_left > 3:
```

```
        return False
```

```
    if (m_left > 0 and m_left < c_left):
```

```
        return False
```

```
    if (m_right > 0 and m_right < c_right):
```

```
        return False
```

```
    return True
```

```

def bfs():
    start = (3, 3, 'L')
    goal = (0, 0, 'R')

    moves = [(1,0),(2,0),(0,1),(0,2),(1,1)]
    queue = deque([(start, [])])
    visited = set()

    while queue:
        state, path = queue.popleft()

        if state in visited:
            continue
        visited.add(state)

        if state == goal:
            return path + [state]

        m, c, boat = state

        for dm, dc in moves:
            if boat == 'L':
                new = (m-dm, c-dc, 'R')
            else:
                new = (m+dm, c+dc, 'L')

            if is_valid(new[0], new[1]):
                queue.append((new, path + [state]))

```

```
solution = bfs()
```

```
print("Solution Path:")
```

```
for step in solution:
```

```
    print(step)
```

Output

```
.py
Solution Path:
(3, 3, 'L')
(3, 1, 'R')
(3, 2, 'L')
(3, 0, 'R')
(3, 1, 'L')
(1, 1, 'R')
(2, 2, 'L')
(0, 2, 'R')
(0, 3, 'L')
(0, 1, 'R')
(1, 1, 'L')
(0, 0, 'R')
```

Result

The **Missionaries and Cannibals Problem** was successfully solved using **Breadth-First Search (BFS)**. The program generated a valid sequence of states to safely transport all missionaries and cannibals across the river without violating the problem constraints.